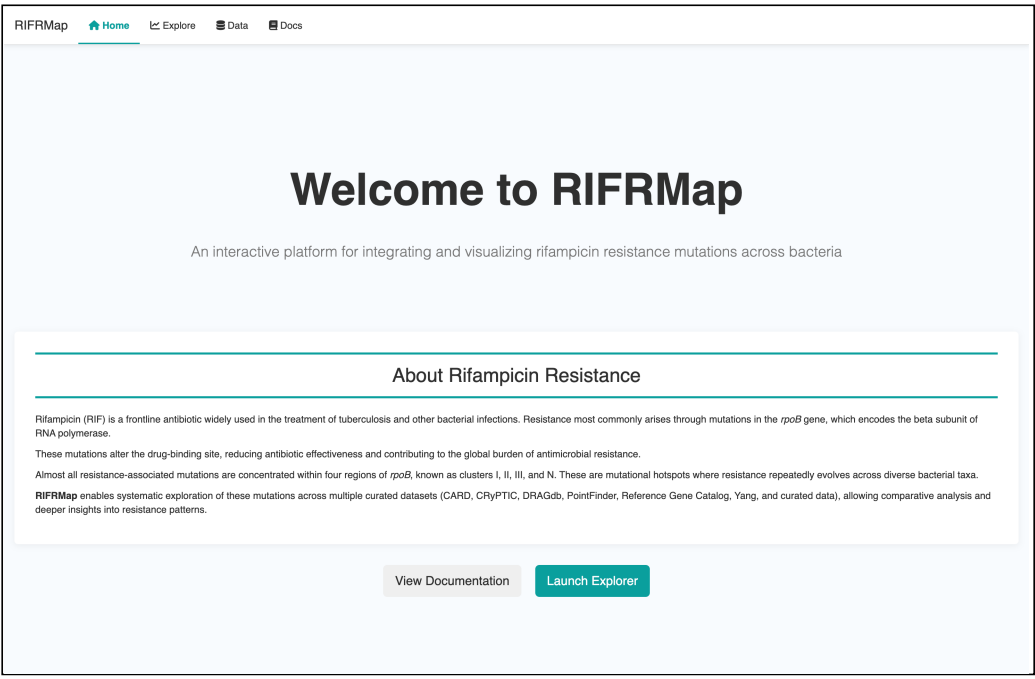
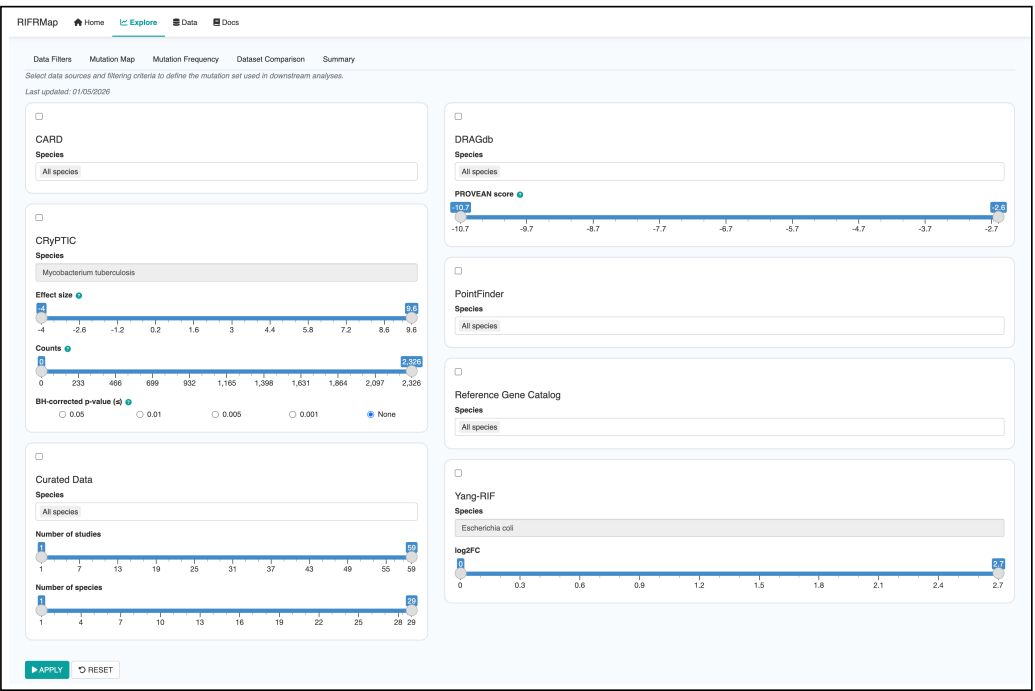
A


RIFRMap
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Welcome to RIFRMap
An interactive platform for integrating and visualizing rifampicin resistance mutations across bacteria
About Rifampicin Resistance
Rifampicin (RIF) is a frontline antibiotic widely used in the treatment of tuberculosis and other bacterial infections. Resistance most commonly arises through mutations in the rpoB gene, which encodes the beta subunit of RNA polymerase.
These mutations alter the drug-binding site, reducing antibiotic effectiveness and contributing to the global burden of antimicrobial resistance.
Almost all resistance-associated mutations are concentrated within four regions of rpoB, known as clusters I, II, III, and N. These are mutational hotspots where resistance repeatedly evolves across diverse bacterial taxa.
RIFRMap enables systematic exploration of these mutations across multiple curated datasets (CARD, CRyPTIC, DRAGdb, PointFinder, Reference Gene Catalog, Yang, and curated data), allowing comparative analysis and deeper insights into resistance patterns.
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Launch Explorer

B


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Data Filters
Mutation Map
Mutation Frequency
Dataset Comparison
Summary
Select data sources and filtering criteria to define the mutation set used in downstream analyses.
Last updated: 01/05/2026
CARD
Species
All species
CRyPTIC
Species
Mycobacterium tuberculosis
Effect size
-4
9.6
-4 -2.6 -1.2 0.2 1.6 3 4.4 5.8 7.2 8.6 9.6
Counts
0
2,326
0 233 466 699 932 1,165 1,398 1,631 1,864 2,097 2,326
BH-corrected p-value (≤)
0.05
0.01
0.005
0.001
None
Curated Data
Species
All species
Number of studies
1
59
1 7 13 19 25 31 37 43 49 55 59
Number of species
1
29
1 4 7 10 13 16 19 22 25 28 29
DRAGdb
Species
All species
PROVEAN score
-10.7
-2.6
-10.7 -9.7 -8.7 -7.7 -6.7 -5.7 -4.7 -3.7 -2.7
PointFinder
Species
All species
Reference Gene Catalog
Species
All species
Yang-RIF
Species
Escherichia coli
log2FC
0
2.7
0 0.3 0.6 0.9 1.2 1.5 1.8 2.1 2.4 2.7
APPLY
RESET